

Consistent spectral recovery of smooth reversible diffusion tensors at a fixed lag

Alec Kriebel
Independent researcher
ORCID: 0009-0001-9320-500X

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Abstract

We give an almost surely consistent spectral-equation estimator for a general smooth symmetric diffusion tensor and an unknown smooth invariant density on a known bounded connected smooth domain. Observations are stationary exact positions at one known fixed positive lag; reflection is conormal, and fixed ellipticity and density bounds are known. A symmetric finite-rank estimate of the stationary pair kernel supplies all its positive empirical eigenpairs. Transition-eigenvalue fourth powers and vanishing ridge regularization give a basis-invariant fit. We prove full spectral-jet excitation and construct the required derivative-consistent kernels from the dependent data. The tensor and its separately fitted divergence converge locally uniformly; an ellipticity-clipped tensor converges globally in L^2 . This provides a convergence analysis for the spectral reconstruction question posed by Reiß in OWR24/2017, in the explicit smooth model stated here. No rate or numerical implementation is claimed. Spectral fitting and eigenvalue-power weighting have established predecessors.

1 Problem, model and contribution

Reiß's contribution with Chorowski, Gobet and Hoffmann asks for convergence analysis of reconstruction of a matrix coefficient from empirical generator eigenpairs and invariant density [1, pp. 1507–1509]. The report leaves the estimator and retained spectral count to be specified. We provide both, for the following class.

Let $D \subset \mathbb{R}^d$, $d \geq 2$, be known, bounded and connected with C^∞ boundary. Fix a known lag $\Delta > 0$. The unknown fields $S = S^\top$ and μ are $C^\infty(\overline{D})$ and satisfy known bounds

$$0 < \ell I \leq S(x) \leq \Lambda I, \quad 0 < c \leq \mu(x) \leq C, \quad \int_D \mu \, dx = 1. \quad (1)$$

Here smoothness on the closed domain means smoothness up to its boundary. The stationary reversible reflected diffusion has generator

$$Lu = \mu^{-1} \operatorname{div}(S \nabla u), \quad n \cdot S \nabla u = 0 \quad \text{on } \partial D. \quad (2)$$

We observe $Y_i = X_{i\Delta}$, $0 \leq i \leq N$, exactly, without observations between these times. The covariance in the interior stochastic differential equation is $\Sigma = 2S/\mu$ and its drift is $\beta = (\operatorname{div} S)/\mu$, where $(\operatorname{div} S)_j = \sum_i \partial_i S_{ij}$. The boundary condition in (2) is essential: ordinary normal reflection cannot be substituted for anisotropic conormal reflection.

Let p_Δ be the transition density relative to Lebesgue measure and $q(x, y) = \mu(x)p_\Delta(x, y)$ the symmetric stationary pair density. The estimator below fits the eigenfunction equations

$$\operatorname{div}(S\nabla u_k) = \nu_k \mu u_k, \quad \kappa_k = e^{\Delta \nu_k} > 0. \quad (3)$$

Its contribution is consistency of this specified empirical spectral fit, including an actual statistical construction and a proved excitation condition. It is pointwise in every fixed (S, μ) satisfying (1), with no known smoothness norm or known spectral gap required for its tuning. We do not establish rough-coefficient or nonreversible results, additional sensor-noise inference, a convergence rate, minimax optimality, computational efficiency, or error control for approximate quadrature and eigensolvers.

Established foundations and priority scope. Fixed-lag identification of a reversible generator is established by Hansen–Scheinkman [2, Proposition 5.5]; multivariate spectral and Dirichlet-form foundations appear in Chen–Hansen–Scheinkman [3]. Crommelin–Vanden-Eijnden already proposed nonparametric drift/tensor spectral residual fitting [4]; their 2011 author manuscript uses powers of transition eigenvalues as weights [5, Eq. (4.7)]. Neither idea is introduced here. Gobet–Hoffmann–Reiß and Chorowski develop scalar one-dimensional spectral inference [6, 7]. Nickl, Giordano–Wang and Alberti–Barnes–Jambhale–Nickl supply multidimensional scalar-conductivity fixed-lag theory [8, 9, 10]. In particular, [10, Lemma 3.8] is a close gradient-excitation predecessor for our cutoff-jet argument and credits earlier work of Alberti. Deterministic matrix inversion from internal solutions has gradient/Hessian and variational predecessors [11, 12].

A dated bounded primary-source audit located no theorem encompassing the particular tensor spectral-consistency result proved here. This is not a worldwide priority guarantee. In particular the full 2011 text inspected was the CWI author manuscript revised August 31, 2011; the publisher-final text was inaccessible, and identity with it is not assumed. Statements about its deferred sample/discretization and infinite-expansion analyses refer to that manuscript [5, Sections 3.2 and 4.2]. Further edition and access limits are recorded in the supplement. Classical nonlinear inverse regularization [15] also yields an alternative estimator-existence argument when its diffusion forward-map hypotheses are supplied; a separately verified derivation of that application accompanies this note. That current derivation is not a located earlier publication and does not establish consistency of the spectral fit studied here. No claim of being the first possible consistent tensor estimator is made.

2 The empirical estimator

We first specify a deterministic smoothing kernel that depends only on D . The notation $C^{a,b}$ denotes the supremum of all mixed derivatives of orders at most a and b in the respective variables.

Lemma 1 (Known-domain smoothing). *There are real, jointly Borel kernels $K_h(x, z)$, $z \in D$, with x in a fixed neighborhood of $\overline{D}$, smooth in x , such that $R_h f(x) = \int_D K_h(x, z) f(z) dz$ converges to f in $C^2(\overline{D})$ for $f \in C^3(\overline{D})$ as $h \downarrow 0$. For each fixed derivative order r ,*

$$\sup_{x,z} |\partial_x^\alpha K_h(x, z)| \leq A_r h^{-d-|\alpha|} \quad (|\alpha| \leq r). \quad (4)$$

On smooth functions of two variables, $R_h \otimes R_h$ converges to the identity in $C^{2,2}(\overline{D} \times \overline{D})$.

Proof. Choose a finite smooth boundary atlas and partition of unity with chart cutoffs. In a flattened chart with inward coordinate $t \geq 0$, extend a cutoff function to a small exterior collar by

$$(Ef)(x', t) = \sum_{a=1}^7 \alpha_a f(x', -at), \quad t < 0, \quad (\alpha_1, \dots, \alpha_7) = (28, -112, 210, -224, 140, -48, 7). \quad (5)$$

The identities $\sum_{a=1}^7 \alpha_a (-a)^k = 1$ for $0 \leq k \leq 6$ match derivatives through order six. Take the collar small enough for all reflected arguments to remain in their charts, and add exterior cutoffs. Interior pieces extend by zero after their cutoff. Summing gives a linear compactly supported extension equal to f on D and bounded in each C^r norm, $r \leq 6$.

Let η be a fixed real $C_c^\infty(\mathbb{R}^d)$ mollifier of integral one and set $R_h f = \eta_h * Ef$ on the indicated neighborhood. In each reflected integral change variables back to $z \in D$. The chart maps have bounded Jacobians bounded away from zero on their cutoff supports; the resulting finite sum is a jointly Borel kernel. Differentiating the mollifier gives (4) for any fixed r . Mollification of a C^3 extension converges in C^2 . Apply the extension independently in both variables to obtain the mixed assertion. The kernels can be signed and need not preserve mass; neither positivity nor exact mass preservation is used. $\square$

Set $K_h(x, z) = 0$ for $z \in \partial D$. This is a Borel convention for every sample in $\bar{D}$; under stationarity, $\mathbb{P}(Y_i \in \partial D) = 0$ for every i , so it changes no almost-sure conclusion.

For $N \geq 2$ put $j = \lfloor \log_2 N \rfloor$, $n = 2^j$ and $h = h_0/(j+1)$, where $h_0 > 0$ is a sufficiently small constant chosen from the known charts. Use the first $n+1$ states and define

$$\tilde{\mu}_j(x) = \frac{1}{n+1} \sum_{i=0}^n K_h(x, Y_i), \quad (6)$$

$$q_j(x, y) = \frac{1}{2n} \sum_{i=0}^{n-1} [K_h(x, Y_i)K_h(y, Y_{i+1}) + K_h(y, Y_i)K_h(x, Y_{i+1})]. \quad (7)$$

Choose a fixed smooth saturation $\psi : \mathbb{R} \rightarrow [c/2, 2C]$ equal to the identity on $[3c/4, 3C/2]$, and set

$$\mu_j = \frac{\psi(\tilde{\mu}_j)}{Z_j}, \quad Z_j = \int_D \psi(\tilde{\mu}_j) dx. \quad (8)$$

In every sample, $\mu_j \geq c/(4C \text{vol}(D))$. It is smooth, positive and normalized. The kernel q_j is real, symmetric and smooth, of rank at most $n+1$, although it can be signed, indefinite, and have a marginal different from μ_j .

On $H = L^2(D, dx)$ define the real self-adjoint finite-rank operator T_j with kernel $q_j(x, y)/\sqrt{\mu_j(x)\mu_j(y)}$. For every positive eigenvalue $\kappa_{j,k}$ choose a real orthonormal basis $\phi_{j,k}$ of its eigenspace and put

$$u_{j,k} = \phi_{j,k}/\sqrt{\mu_j}, \quad \nu_{j,k} = \Delta^{-1} \log \kappa_{j,k}. \quad (9)$$

All positive eigenvalues are retained; negative and zero ones are omitted. Nonzero modes are smooth because they lie in the span of the smooth kernel sections. No labeling relative to population eigenvectors is imposed.

For a smooth u define the column jet and matching unknown vector

$$\begin{aligned} Ju &= ((\partial_{ii}u)_i, (2\partial_{ij}u)_{i<j}, (\partial_i u)_i)^\top, \\ \theta &= ((S_{ii})_i, (S_{ij})_{i<j}, (\text{div } S)_i)^\top \in \mathbb{R}^m, \quad m = d(d+1)/2 + d. \end{aligned} \quad (10)$$

Thus $(Ju)^\top \theta = \operatorname{div}(S \nabla u)$. For each $x \in D$ let

$$\hat{\theta}_j(x) = \arg \min_{z \in \mathbb{R}^m} \left\{ \sum_{\kappa_{j,k} > 0} \kappa_{j,k}^4 [(Ju_{j,k}(x))^\top z - \nu_{j,k} \mu_j(x) u_{j,k}(x)]^2 + \lambda_j |z|^2 \right\}, \quad \lambda_j = (j+1)^{-1}. \quad (11)$$

The minimizer is unique even if there are no positive modes. Write its tensor part as the symmetric matrix $\hat{S}_j$, and its remaining part as $\hat{v}_j$. The latter estimates $\operatorname{div} S$ separately; it is not the divergence of $\hat{S}_j$. Let $\hat{S}_j^c$ be the pointwise Frobenius projection of $\hat{S}_j$ onto $\{M = M^\top : \ell \mathbf{I} \leq M \leq \Lambda \mathbf{I}\}$. At sample size N , use these estimators indexed by j ; any fixed admissible tensor may be used for the finitely many smaller sample sizes.

Theorem 2 (Fixed-lag tensor consistency). *Under (1)–(2) and the stationary exact observation model above, the specified estimators are measurable and, on one probability-one event, for every compact $E \subseteq D$,*

$$\|\hat{S}_j - S\|_{C(E)} + \|\hat{v}_j - \operatorname{div} S\|_{C(E)} \longrightarrow 0, \quad \|\hat{S}_j^c - S\|_{L^2(D)} \longrightarrow 0. \quad (12)$$

Also $\mu_j \rightarrow \mu$ in $C^2(\overline{D})$, so $2\hat{S}_j/\mu_j$ and $\hat{v}_j/\mu_j$ recover the covariance and drift locally uniformly. The limits hold along all $N \rightarrow \infty$ by the dyadic prefix convention. Any positive deterministic $\lambda_j \rightarrow 0$ in (11) gives the same consistency conclusion.

3 Derivative convergence from the dependent observations

We give the statistical part explicitly, including overlapping pairs. The form of $-L$ on $L^2(\mu dx)$ is $\mathcal{E}(f, f) = \int_D \nabla f^\top S \nabla f dx$. If $\int_D f \mu dx = 0$, the Neumann Poincaré constant C_D gives

$$\int_D f^2 \mu dx = \min_a \int_D (f - a)^2 \mu dx \leq C \min_a \int_D (f - a)^2 dx \leq CC_D \int_D |\nabla f|^2 dx.$$

Consequently $P = e^{\Delta L}$ contracts mean-zero functions by $\rho \leq \exp[-\Delta \ell / (CC_D)] < 1$.

For a bounded pair observable $F(Y_i, Y_{i+1})$, center it as Z_i . Let $r(z) = \mathbb{E}[Z_0 \mid Y_1 = z]$ and $s(z) = \mathbb{E}[Z_0 \mid Y_0 = z]$. They have mean zero and $L^2(\mu)$ norms at most $\sqrt{\operatorname{Var} F}$. Stationarity and the Markov property, conditioning at the endpoints, give

$$\operatorname{Cov}(Z_0, Z_a) = \langle r, P^{a-1} s \rangle_{L^2(\mu)}, \quad a \geq 1. \quad (13)$$

This includes the overlapping case $a = 1$. Hence $|\operatorname{Cov}(Z_0, Z_a)| \leq \rho^{a-1} \operatorname{Var} F$, and

$$\operatorname{Var} \left(\frac{1}{n} \sum_{i=0}^{n-1} F(Y_i, Y_{i+1}) \right) \leq \frac{1 + 2/(1 - \rho)}{n} \|F\|_\infty^2. \quad (14)$$

The analogous state-observable bound follows with P^a . These inequalities apply separately to each bandwidth-dependent observable; independence of pairs is never assumed.

Lemma 3 (Population regularity and empirical limits). *The kernel q is smooth on $\overline{D} \times \overline{D}$. Almost surely,*

$$\tilde{\mu}_j \longrightarrow \mu \text{ in } C^2(\overline{D}), \quad q_j \longrightarrow q \text{ in } C^{2,2}(\overline{D} \times \overline{D}), \quad \mu_j \longrightarrow \mu \text{ in } C^2(\overline{D}). \quad (15)$$

Proof. The smooth elliptic conormal realization has compact resolvent. If $-Lu_k = \gamma_k u_k$, the min–max principle and (1) compare γ_k with the Neumann Laplacian eigenvalues; their counting is polynomial. Smooth elliptic graph regularity and Sobolev embedding give

$$\|f\|_{H^{2a}(D)} \leq B_a \|(I - L)^a f\|_{L^2(\mu)}, \quad \|u_k\|_{C^r(\overline{D})} \leq B_r(1 + \gamma_k)^{b_r} \quad (16)$$

for sufficiently high integer a and some finite b_r . The graph estimates are the standard smooth conormal elliptic estimates; see [13, Theorem 2.2 and Corollary 2.3] and the Sobolev embedding [14, Corollary 6.12]. The weighted and ordinary L^2 norms are equivalent; use the positive shift $I - L$ to include the constant mode. Thus every fixed differentiated series

$$q(x, y) = \mu(x)\mu(y) \sum_k e^{-\Delta\gamma_k} u_k(x)u_k(y)$$

converges absolutely uniformly. This proves the asserted positive-time regularity, using population elliptic theory rather than estimated modes.

Reversibility and stationarity give $\mathbb{E}\tilde{\mu}_j = R_h\mu$ and $\mathbb{E}q_j = (R_h \otimes R_h)q$; their deterministic biases vanish in the displayed norms by Lemma 1. For $|\alpha|, |\beta| \leq 2$, each differentiated summand of (7) is bounded by a constant times $(j+1)^{2d+4}$. By (14) its average has variance at each (x, y) at most $B(j+1)^{4d+8}2^{-j}$. A deterministic net of $\overline{D} \times \overline{D}$ with spacing $\varepsilon_j = 2^{-j/(4d)}$ has at most $B2^{j/2}$ points. Chebyshev’s inequality and a union bound yield

$$\mathbb{P}\left(\max_{\text{net}} |\partial_x^\alpha \partial_y^\beta (q_j - \mathbb{E}q_j)| > (j+1)^{-1}\right) \leq B(j+1)^{4d+10}2^{-j/2}. \quad (17)$$

The bound is summable. The derivatives in question, and their expectations, are Lipschitz on the fixed neighborhood with bound $B(j+1)^{2d+5}$, by one further derivative in (4). Interpolation from the net costs at most $B(j+1)^{2d+5}\varepsilon_j \rightarrow 0$. The neighborhood makes the short line segments legitimate even if D is not convex. Borel–Cantelli gives uniform convergence of the centered derivatives.

For (6), the state-observable bound, derivative bound $B(j+1)^{d+2}$ and a d -dimensional net with the same spacing give the summable probability bound $B(j+1)^{2d+6}2^{-3j/4}$ at threshold $(j+1)^{-1}$. There are only finitely many derivative indices, so all the asserted limits hold on one event of probability one. Uniform convergence eventually places $\tilde{\mu}_j$ in $[3c/4, 3C/2]$, making saturation inactive. Then $Z_j \rightarrow \int_D \mu dx = 1$, proving the last limit. No mixing parameter or truth-dependent regularity constant was used in the tuning sequence. $\square$

4 Basis-invariant deterministic reconstruction

The argument below also explains why a fixed number of modes or individual eigenvector consistency is unnecessary. On $H = L^2(D, dx)$, let T have kernel $q(x, y)/\sqrt{\mu(x)\mu(y)}$. Multiplication by $\sqrt{\mu}$ conjugates P to T ; its orthonormal eigenfunctions are $\phi_k = \sqrt{\mu}u_k$, with eigenvalues $\kappa_k = e^{-\Delta\gamma_k} > 0$, including the stationary constant eigenfunction. Zero is only an accumulation point. Put $M_j = \mu_j^{-1/2}$, $M = \mu^{-1/2}$ and $W_j = M_j T_j$, $W = MT$.

Fix $E \Subset D$. Lemma 3, the common density lower bound, and Cauchy–Schwarz in the kernel’s second variable imply

$$\|T_j - T\|_{\text{op}} \longrightarrow 0, \quad \|W_j - W\|_{H \rightarrow C^2(E)} \longrightarrow 0. \quad (18)$$

For clarity, the needed deterministic hypotheses are merely uniform convergence of μ_j , local C^2 convergence of μ_j , global L^2 convergence of q_j , and $\sup_{x \in E} \|\partial_x^\alpha (q_j - q)(x, \cdot)\|_{L^2(D)} \rightarrow 0$ for $|\alpha| \leq 2$.

Multiplication and its derivatives are controlled by the positive lower bound. The global kernel bound gives Hilbert–Schmidt, hence operator, convergence of T_j ; differentiating the W_j kernel gives the second assertion of (18).

Define bounded operators $B_j(x) : H \rightarrow \mathbb{R}^m$ and $e_j(x) : H \rightarrow \mathbb{R}$ by $B_j(x)f = J(W_j f)(x)$ and $e_j(x)f = (W_j f)(x)$, and similarly B, e . Their norms converge uniformly on E . Set

$$g(t) = \begin{cases} t^2 & t > 0, \\ 0 & t \leq 0, \end{cases} \quad h(t) = \begin{cases} t^2 \log t & t > 0, \\ 0 & t \leq 0. \end{cases} \quad (19)$$

Both are continuous at zero. The normal equation for (11) is

$$\hat{\theta}_j = (A_j + \lambda_j \mathbf{I})^{-1} r_j, \quad A_j = B_j g(T_j) B_j^*, \quad r_j = \frac{\mu_j}{\Delta} B_j h(T_j) e_j^*. \quad (20)$$

Indeed $W_j \phi_{j,k} = \kappa_{j,k} u_{j,k}$, so the outer factors supply two eigenvalue powers and the middle function supplies two more. These identities apply to the entire finite spectrum, including negative modes, without taking their logarithms. They are invariant under orthogonal basis changes within repeated eigenspaces.

The spectra lie eventually in a common bounded interval by (18). Polynomial approximation of continuous functions on that interval, and continuity of each polynomial in a bounded operator, show that $g(T_j) \rightarrow g(T)$ and $h(T_j) \rightarrow h(T)$ in operator norm. Therefore

$$A_j \longrightarrow A = Bg(T)B^*, \quad r_j \longrightarrow r = (\mu/\Delta)Bh(T)e^* \quad \text{uniformly on } E. \quad (21)$$

This includes empirical eigenvalue splits, crossings through zero and small positive eigenvalues. Initial eigenvalues greater than one cause no problem and are not interpreted as population generator eigenvalues.

Proposition 4 (Full spectral-jet excitation). *For every $x \in D$, $A(x)$ is positive definite and $r(x) = A(x)\theta(x)$. For each compact $E \subseteq D$ there is a truth-dependent $a_E > 0$ with $A(x) \geq a_E \mathbf{I}$ on E .*

Proof. Spectral expansion of the bounded-functional formulas gives

$$A(x) = \sum_k \kappa_k^4 J u_k(x) J u_k(x)^\top, \quad r(x) = \sum_k \kappa_k^4 \nu_k \mu(x) u_k(x) J u_k(x).$$

These series converge by the smoothing already proved (or by Parseval for the bounded functionals). Equation (3) gives $r = A\theta$. If $\xi^\top A(x)\xi = 0$, each nonnegative term vanishes and $\xi^\top J u_k(x) = 0$ for every k , since all $\kappa_k > 0$. Every $f \in C_c^\infty(D)$ belongs to every power domain of $\mathbf{I} - L$: successive applications of the local differential operator retain interior support and satisfy all iterated conormal conditions. Its spectral partial sums converge in each graph norm. Equation (16) and Sobolev embedding, choosing $2a > d/2 + 2$, imply convergence in $C^2(\overline{D})$. Thus $\xi^\top J f(x) = 0$ for every such f .

Multiply affine and quadratic polynomials centered at x by a cutoff equal to one near x . Their gradients and symmetric Hessians can be prescribed arbitrarily; the resulting jets span $\mathbb{R}^m$. Hence $\xi = 0$. Finally $A(x)$ is continuous by continuity of the evaluation functionals; compactness of E gives $a_E > 0$. This proves excitation rather than assuming an unverified rank condition. No class-uniform conditioning bound is claimed. $\square$

Consistency in Theorem 2. Work on the event of Lemma 3. By (21), eventually $A_j \geq a_E \mathbf{I}/2$ uniformly on E . The exact identity

$$\hat{\theta}_j - \theta = (A_j + \lambda_j \mathbf{I})^{-1} [(r_j - r) + (A - A_j)\theta - \lambda_j \theta]$$

proves local uniform convergence for any positive $\lambda_j \rightarrow 0$. No relative noise/ridge rate is needed because the limiting local matrix is positive definite. This also proves the stated local drift and covariance limits, using the density lower bound.

Projection onto the closed convex matrix interval in Frobenius norm is nonexpansive and fixes S . On a countable compact exhaustion of D , the projected tensor converges pointwise in the interior and is uniformly bounded by the known matrix interval. Since D has finite volume, dominated convergence gives the global L^2 limit. There is no corresponding global claim about the fitted divergence or derivatives of the tensor estimate. All sample sizes use a dyadic index tending to infinity. Measurability is proved next, completing the theorem. $\square$

5 Finite-feature measurability and verification

The eigenbasis in (9) need not be chosen measurably. The following equivalent formulas avoid it. For $0 \leq i \leq n$ put $f_i = K_h(\cdot, Y_i)/\sqrt{\mu_j}$ and $v_i = K_h(\cdot, Y_i)/\mu_j$. Let $V : \mathbb{R}^{n+1} \rightarrow H$ have columns f_i , and let C_n be the symmetric matrix with adjacent entries $(C_n)_{i,i+1} = (C_n)_{i+1,i} = 1/(2n)$ and all others zero. Then

$$T_j = VC_nV^*, \quad G = V^*V \geq 0, \quad H_n = G^{1/2}C_nG^{1/2}.$$

Let D_x have columns $Jv_i(x)$ and let $v(x)$ be the column of $v_i(x)$. The formulas in (20) become

$$\begin{aligned} A_j(x) &= D_x C_n G^{1/2} g(H_n) G^{1/2} C_n D_x^\top, \\ r_j(x) &= \frac{\mu_j(x)}{\Delta} D_x C_n G^{1/2} h(H_n) G^{1/2} C_n v(x). \end{aligned} \tag{22}$$

To check them, $B_j = D_x C_n V^*$ and $e_j = v(x)^\top C_n V^*$, and

$$V^* a(V C_n V^*) V = G^{1/2} a(H_n) G^{1/2}$$

holds first for polynomials a and then for continuous a by uniform approximation. No inverse of G is used, so singular feature Gram matrices are allowed. Kernel sections and their derivatives are Borel functions of the data; known-domain integrals defining Z_j and G are Borel as well. Positive square root and continuous functional calculus of finite symmetric matrices are continuous, and $A_j + \lambda_j I$ is invertible. Thus (22) proves joint measurability and continuity in x . Separability on compact sets, and the bounded projected output, give the usual $C(E)$ and $L^2(D)$ random-element measurability. Multiplicities, rank loss and zero crossings require no eigenvector selection.

The supplement contains the source, a portable finite-control runner, expected scientific outputs, exact source-edition notes, and the separately verified classical regularization application. Its controls check algebra, signed-kernel moments, covariance identities, spectral collisions and boundary/model distinctions. They supplement the analytic proof and do not prove an infinite-dimensional theorem, simulate the estimator's asymptotic performance, or supply a general-domain numerical implementation.

AI use and review status. AI tools were used extensively in deriving, checking and writing this work, including separate adversarial mathematical and priority audits. Their checks are disclosed as AI-assisted verification, not independent human peer review or a proof-assistant certificate. This preprint is unrefereed and has not undergone human peer review. The author takes responsibility for the claims and for the explicit scope and source-access limitations.

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